

Exercise J-2

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON EXPANSION, PROPERTIES OF DETERMINANT

1. Without expanding as far as possible, prove that :

जहाँ तक संभव हो बिना विस्तार किये सिद्ध कीजिए कि :

$$(a) \begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ 2a + 1 & a + 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a - 1)^3$$

$$(b) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$$

Ans.

Sol. (a) $\begin{vmatrix} a^2 + 2a & 2a + 1 & 1 \\ 2a + 1 & a + 2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a - 1)^3$

$$C_2 \rightarrow C_2 - C_1$$

$$\begin{vmatrix} a^2 + 2a & 1 + a & 1 \\ 2a + 1 & 1 - a & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$$(1 - a) \begin{vmatrix} a^2 + 2a & 1 + a & 1 \\ 2a + 1 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - 2C_2$$

$$(1 - a) \begin{vmatrix} a^2 - 2 & 1 + a & 1 \\ 2a - 1 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$$(1 - a) (a^2 - 2(1 - 0) - (1 + 0)(2a - 1 - 3) + 1(0 - 3))$$

$$(1 - a) (a^2 - 2 - (1 + a)(2a - 4) - 3)$$

$$(1 - a) (a^2 - 2 - 2a + 4 - 2a^2 + 4a - 3)$$

$$(1 - a) (-a^2 + 2a - 1)$$

$$-(1 - a) (a^2 - 2a + 1)$$

$$(1-a)(a-1)^2 \\ \Rightarrow (a-1)^3$$

$$(b) \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

$$C_2 \rightarrow C_2 - C_1 \quad C_3 \rightarrow C_3 - C_1$$

2. If $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

यदि $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ तथा x, y, z सभी भिन्न है तो सिद्ध कीजिए $xyz = -1$

Ans.

Sol. $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$

$$\begin{vmatrix} x^3 & x^2 & x \\ y^3 & y^2 & y \\ z^3 & z^2 & z \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix}$$

$$xyz \begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} = 0$$

$$xyz \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} (xyz + 1) = 0$$

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (xyz + 1) = 0$$

x, y, z are all diff. then $xyz + 1 = 0 \Rightarrow xyz = -1$

3. Using properties of determinants or otherwise evaluate $\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$.

$$\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix} \text{ का मान ज्ञात कीजिए।}$$

Ans. -1

Sol. $\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 2R_1$$

$$\begin{vmatrix} 18 & 40 & 89 \\ 4 & 9 & 20 \\ 9 & 20 & 44 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - (2R_2 + R_3)$$

$$= \begin{vmatrix} 1 & 2 & 5 \\ 4 & 9 & 20 \\ 9 & 20 & 44 \end{vmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{vmatrix} 1 & 2 & 5 \\ 4 & 9 & 20 \\ 1 & 2 & 4 \end{vmatrix}$$

$$1(36 - 40) - 2(16 - 20) + 5(8 - 9)$$

$$-4 + 8 - 5 = -1$$

4. Prove that (सिद्ध कीजिए) $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$.

Ans.

Sol. $R_1 \rightarrow R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1 \quad C_3 \rightarrow C_3 - C_1$$

$$(a+b+c) \begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

$$(a+b+c)^3 \begin{vmatrix} 1 & 0 & 0 \\ 2b & -1 & 0 \\ 2c & 0 & -1 \end{vmatrix} \Rightarrow (a+b+c)^3$$

5. Prove that for all values of θ ,
$$\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} = 0$$

सिद्ध कीजिए कि θ के सभी मानों के लिए,
$$\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} = 0 \text{ होगा।}$$

Ans.

Sol. $R_2 \rightarrow R_2 + R_3$

$$\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ 2\sin \theta \cos \frac{2\pi}{3} & 2\cos \theta \cos \frac{2\pi}{3} & 2\sin 2\theta \cos \frac{4\pi}{3} \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ -\sin \theta & -\cos \theta & -\sin 2\theta \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$$

0

6. Prove that (सिद्ध कीजिए)
$$\begin{vmatrix} (\beta + \gamma - \alpha - \delta)^4 & (\beta + \gamma - \alpha - \delta)^2 & 1 \\ (\gamma + \alpha - \beta - \delta)^4 & (\gamma + \alpha - \beta - \delta)^2 & 1 \\ (\alpha + \beta - \gamma - \delta)^4 & (\alpha + \beta - \gamma - \delta)^2 & 1 \end{vmatrix} = -64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$$

Ans.

Sol. Let $a = \beta + \gamma - \delta - \alpha$, $b = \gamma + \alpha - \beta - \delta$, $c = \alpha + \beta - \gamma - \delta$

we get
$$\begin{vmatrix} a^4 & a^2 & 1 \\ b^4 & b^2 & 1 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$$

$$= \begin{vmatrix} a^4 - b^4 & a^2 - b^2 & 0 \\ b^4 - c^4 & b^2 - c^2 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$= (a^2 - b^2)(b^2 - c^2) \begin{vmatrix} a^2 + b^2 & 1 & 0 \\ b^2 + c^2 & 1 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$$

$$= (a^2 - b^2)(b^2 - c^2)(a^2 - c^2)$$

$$= (a - b)(a + b)(b - c)(b + c)(a + c)(a - c)$$

$$= -2^6(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma)(\alpha - \delta)(\beta - \delta)(\gamma - \delta)$$

7. Solve for x the equation
$$\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0.$$

x के लिये निम्न समीकरण को सरल कीजिए
$$\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0.$$

Ans. $x = n\pi, n \in I$

Sol. already done

8. Solve for x

ग के लिये हल कीजिए

$$(a) \begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$$

$$(b) \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

Ans. (a) $x = -1$ or $x = -2$; (b) $x = 4$

Sol. (a) $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ -1 & -x-2 & -2x-3 \\ -1 & -x-1 & x+5 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 0 & -1 & -3x-8 \\ -1 & -x-1 & x+5 \end{vmatrix}$$

$$(x+1)(x^2+3x+2)=0$$

$$x = -1, -2$$

$$(b) \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 - R_2$$

$$R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} 2 & 6 & 12 \\ 4 & 18 & 48 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

$$R_3 \rightarrow R_3 - (R_1 + R_2)$$

$$\begin{vmatrix} 2 & 6 & 12 \\ 4 & 18 & 48 \\ 2x-4 & 2x-3 & 3x-4 \end{vmatrix} = 0$$

$$\Rightarrow x = 4$$

9. If $a+b+c=0$, solve for x : $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$.

यदि $a+b+c=0$ है, तो x के लिये हल कीजिये : $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$.

Ans. $x = 0$ or $x = \pm \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$

Sol. Hint : On applying $R_1 \rightarrow R_1 + R_2 + R_3$
and then solving the determinant we get

$$x = 0 \text{ or } x^2 = a^2 + b^2 + c^2 - ab - bc - ca$$

$$x = \pm \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$$

10. If $p+q+r=0$, prove that $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$.

यदि $p+q+r=0$, सिद्ध कीजिये $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

Ans.

Sol.
$$\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$$

$$\Rightarrow pa(a^2qr - p^2bc) - qb(q^2ac - prb^2) + rc(pqc^2 - r^2ab)$$

$$\Rightarrow a^3pqr - p^3abc - q^3abc + b^3pqr + c^3pqr - r^3abc$$

$$\Rightarrow pqr(a^3 + b^3 + c^3) - abc(p^3 + q^3 + r^3)$$

$$pqr(a^3 + b^3 + c^3) - abc(3pqr)$$

$$(pqr)(a^3 + b^3 + c^3 - 3abc)$$

$$-pqr \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

11. If a, b, c are all different and $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$, then prove that, $abc(ab+bc+ca) = a+b+c$.

यदि a, b, c सभी भिन्न हैं तथा $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$, तो सिद्ध कीजिए $abc(ab+bc+ca) = a+b+c$

Ans.

Sol.
$$\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} - \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix}$$

$$(abc) \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} - \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = 0$$

$$abc(a-b)(b-c)(c-a)(ab+bc+ca) - (a-b)(b-c)(c-a)(a+b+c) = 0$$

$$(a-b)(b-c)(c-a)\{abc(ab+bc+ca) - (a+b+c)\} = 0$$

a, b, c are all diff. so,

$$abc(ab+bc+ca) = a+b+c$$

12. Show that, $\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix}$ is divisible by λ^2 and find the other factor.

दर्शाये कि सारणिक, $\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix}$, λ^2 से भाज्य है तथा इसका अन्य गुणनखण्ड भी ज्ञात कीजिए।

Ans. $\lambda^2 (a^2 + b^2 + c^2 + \lambda)$

Sol. $(abc) \begin{vmatrix} a + \frac{\alpha}{a} & a & a \\ b & b + \frac{\lambda}{b} & b \\ c & c & c + \frac{\lambda}{c} \end{vmatrix}$

$$\begin{vmatrix} a^2 + \lambda & a^2 & a^2 \\ b^2 & b^2 + \lambda & b^2 \\ c^2 & c^2 & c^2 + \lambda \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{vmatrix} \lambda + a^2 + b^2 + c^2 & \lambda^2 + a^2 + b^2 + c^2 & \lambda + a^2 + b^2 + c^2 \\ b^2 & b^2 + \lambda & b^2 \\ c^2 & c^2 & c^2 + \lambda \end{vmatrix}$$

$$\Rightarrow (\lambda + a^2 + b^2 + c^2) \begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2 + \lambda & b^2 \\ c^2 & c^2 & c^2 + \lambda \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1 \quad C_3 \rightarrow C_3 - C_1$$

$$\Rightarrow (\lambda + a^2 + b^2 + c^2) \begin{vmatrix} 1 & 0 & 0 \\ b^2 & \lambda & 0 \\ c^2 & 0 & \lambda \end{vmatrix}$$

$$\Rightarrow (\lambda + a^2 + b^2 + c^2)(\lambda^2)$$

13. Evaluate $\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$

$$\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix} \text{ मान ज्ञात कीजिए।}$$

Ans. $5(3\sqrt{2} - 5\sqrt{3})$

Sol. $\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix} = \begin{vmatrix} \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} & 5 & \sqrt{10} \\ 3 & \sqrt{15} & 5 \end{vmatrix} + \begin{vmatrix} \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{26} & 5 & \sqrt{10} \\ \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$

$$= \sqrt{3}\sqrt{5}\sqrt{5} \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{vmatrix} + \sqrt{13}\sqrt{5}\sqrt{5} \begin{vmatrix} 1 & 2 & 1 \\ \sqrt{2} & \sqrt{5} & \sqrt{2} \\ \sqrt{5} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_2$$

$$= 5\sqrt{3} \begin{vmatrix} -1 & 2 & 1 \\ 0 & \sqrt{5} & \sqrt{2} \\ 0 & \sqrt{3} & \sqrt{5} \end{vmatrix} = 5(3\sqrt{2} - 5\sqrt{3})$$

14. Let $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$. Find the sum $\sum_{r=1}^n \Delta_r$

माना $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$ योग $\sum_{r=1}^n \Delta_r$ का मान ज्ञात कीजिए।

Ans. 0

Sol.
$$\begin{vmatrix} \sum_{r=1}^n r-1 & n & 6 \\ \sum_{r=1}^n (r-1)^2 & 2n^2 & 4n-2 \\ \sum_{r=1}^n (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\begin{vmatrix} \frac{n(n-1)}{2} & n & 6 \\ \frac{n(n-1)(2n-1)}{6} & 2n^2 & 4n-2 \\ \left(\frac{n(n-1)}{2}\right)^2 & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\frac{n(n-1)}{2} \begin{vmatrix} 1 & n & 6 \\ \frac{2n-1}{3} & 2n^2 & 4n-2 \\ \frac{n(n-1)}{2} & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\frac{n(n-1)}{2 \times 6} \begin{vmatrix} 6 & n & 6 \\ (2n-1)2 & 2n^2 & 4n-2 \\ 3n(n-1) & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$C_1 = C_3$$

MULTIPLICATION OF DETERMINANT

15. Prove that (सिद्ध कीजिए) $\begin{vmatrix} (a_1 - b_1)^2 & (a_1 - b_2)^2 & (a_1 - b_3)^2 \\ (a_2 - b_1)^2 & (a_2 - b_2)^2 & (a_2 - b_3)^2 \\ (a_3 - b_1)^2 & (a_3 - b_2)^2 & (a_3 - b_3)^2 \end{vmatrix} = 2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$

Ans.

Sol. $\begin{vmatrix} a_1^2 - 2a_1b_1 + b_1^2 & a_1^2 - 2a_1b_2 + b_2^2 & a_1^2 + b_2^2 - 2a_1b_3 \\ a_2^2 + b_1^2 - 2a_2b_1 & a_2^2 + b_2^2 - 2a_2b_2 & a_2^2 + b_3^2 - 2a_2b_3 \\ a_3^2 + b_1^2 - 2a_3b_1 & a_3^2 + b_2^2 - 2a_3b_2 & a_3^2 + b_3^2 - 2a_3b_3 \end{vmatrix}$

$$\begin{vmatrix} a_1^2 & -2a_1 & 1 \\ a_2^2 & -2a_2 & 1 \\ a_3^2 & -2a_3 & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ b_1 & b_2 & b_3 \\ b_1^2 & b_2^2 & b_3^2 \end{vmatrix}$$

$$2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$$

16. Prove that (सिद्ध कीजिए) $\begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha\beta + \gamma\delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \beta)(\gamma + \delta) & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \\ \alpha\beta + \gamma\delta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) & 2\alpha\beta\gamma\delta \end{vmatrix} = 0.$

Ans.

Sol. $\begin{vmatrix} 1+1 & (\alpha + \beta) + (\gamma + \delta) & \alpha\beta + \gamma\delta \\ (\alpha + \beta) + (\gamma + \delta) & 2(\alpha + \beta)(\gamma + \delta) & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \\ \alpha\beta + \gamma\delta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) & 2\alpha\beta\gamma\delta \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 0 \\ (\gamma + \delta) & \alpha + \beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix} \begin{vmatrix} 1 & \alpha + \beta & \alpha\beta \\ 1 & \gamma + \delta & \gamma\delta \\ 0 & 1 & 1 \end{vmatrix}$$

17. Prove that (सिद्ध कीजिए) :
$$\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$$

Ans.

Sol.
$$\begin{vmatrix} a^2 + p^2 - 2ap & a^2 + q^2 - 2aq & a^2 + r^2 - 2ar \\ b^2 + p^2 - 2bp & b^2 + q^2 - 2bq & b^2 + r^2 - 2br \\ c^2 + p^2 - 2pc & c^2 + q^2 - 2cq & c^2 + r^2 - 2cr \end{vmatrix}$$

$$\begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix}$$

$$- \begin{vmatrix} 1 & -2a & a^2 \\ 1 & -2b & b^2 \\ 1 & -2c & c^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix}$$

$$\begin{vmatrix} 1 + 2ap + a^2p^2 & 1 + 2aq + a^2q^2 & 1 + 2ar + a^2r^2 \\ - & - & - \\ - & - & - \end{vmatrix}$$

$$= \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$$

18. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$.

यदि $S_r = \alpha^r + \beta^r + \gamma^r$ हो, तो दर्शाइये कि $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$

Ans.

Sol. $\begin{vmatrix} 3 & \alpha + \beta + \gamma & \alpha^2 + \beta^2 + \gamma^2 \\ \alpha + \beta + \gamma & \alpha^2 + \beta^2 + \gamma^2 & \alpha^3 + \beta^3 + \gamma^3 \\ \alpha^2 + \beta^2 + \gamma^2 & \alpha^3 + \beta^3 + \gamma^3 & \alpha^4 + \beta^4 + \gamma^4 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \times \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$$

$$= (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$$

SYSTEM OF EQUATIONS

19. Find the real values of r for which the following system of linear equations has a non-trivial solution. Also find the non-trivial solutions :

$$2rx - 2y + 3z = 0$$

$$x + ry + 2z = 0$$

$$2x + rz = 0$$

r के किस वास्तविक मान के लिये रेखीय समीकरण निकाय :

$$2rx - 2y + 3z = 0$$

$$x + ry + 2z = 0$$

$$2x + rz = 0 \quad \text{के अशून्य हल होंगे।}$$

Ans. $r = 2$; $x = k$; $y = k/2$; $z = -k$ where $k \in \mathbb{R} - \{0\}$

Sol. $D = \begin{vmatrix} 2r & -2 & 3 \\ 1 & r & 2 \\ 2 & 0 & r \end{vmatrix} = 0$

$$2r(r^2 - 0) + 2(r - 4) + 3(0 - 2r) = 0$$

$$2r^3 + 2r - 8 - 6r = 0$$

$$2r^3 - 4r - 8 = 0$$

$$r^3 - 2r - 4 = 0$$

$$r^2(r-2) + 2r(r-2) + 2(r-2)$$

$$(r-2)(r^2 + 2r + 2) = 0$$

$$(r-2)$$

$$r = 2$$

$$\text{Let } x = k$$

$$2k + 2z = 0$$

$$z = -k$$

$$k + 2y - 2k = 0$$

$$2y = k$$

$$y = \frac{k}{2}$$

non trivial solution

$$\left(k, \frac{k}{2}, -k\right)$$

20. Solve the following using Cramer's rule and state whether consistent or not :

क्रमेर के नियम का प्रयोग करते हुए निम्न समीकरण निकायों को हल कीजिये तथा बताइयें कि यह संगत हैं या नहीं :

$$\begin{array}{lll} x + y + z - 6 = 0 & x + 2y + z = 1 & 7x - 7y + 5z = 3 \\ \text{(a) } 2x + y - z - 1 = 0 & \text{(b) } 3x + y + z = 6 & \text{(c) } 3x + y + 5z = 7 \\ x + y - 2z + 3 = 0 & x + 2y = 0 & 2x + 3y + 5z = 5 \end{array}$$

Ans. (a) $x = 1, y = 2, z = 3$; consistent (b) $x = 2, y = -1, z = 1$; consistent, (c) inconsistent

Sol. (a) $D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 3$

$$\Rightarrow D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 1 & 1 & -1 \\ -3 & 1 & -2 \end{vmatrix} = 3$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 1 & -1 \\ 1 & -3 & -2 \end{vmatrix} = 6$$

Similarly $D_3 = 9$

so consistent having $x = 1, y = 2, z = 3$

$$(c) D = \begin{vmatrix} 7 & -7 & 5 \\ 3 & 1 & 5 \\ 2 & 3 & 5 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} 3 & -7 & 1 \\ 7 & 1 & 1 \\ 5 & 3 & 1 \end{vmatrix} = 24$$

Inconsistent system

21. For what value of K do the following system of equations possess a non trivial (i.e. not all zero) solution over the set of rationals Q ?

$$x + Ky + 3z = 0, 3x + Ky - 2z = 0, 2x + 3y - 4z = 0.$$

For that value of K , find all the solutions of the system.

झ के किन मानों के लिये निम्नलिखित समीकरणों के निकाय का परिमेय संख्याओं के समुच्चय Q में अनिरर्थक हल (अर्थात् सभी शून्य) हल है।

$$x + Ky + 3z = 0, 3x + Ky - 2z = 0, 2x + 3y - 4z = 0.$$

झ के उस मान के लिये निकाय के सभी हल भी ज्ञात कीजिये।

Ans. $K = \frac{33}{2}, x : y : z = -\frac{15}{2} : 1 : -3$

Sol. $D = \begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$

$$1(-4k + 6) - k(-12 + 4) + 3(9 - 2k) = 0$$

$$-4k + 6 + 12k - 4k + 27 - 6k = 0$$

$$33 - 2k = 0$$

$$k = \frac{33}{2}$$

Let $x = k$

$$ky + 3z = -k$$

$$ky - 2z = -3k$$

$$- \quad + \quad +$$

$$5z = 2k$$

$$z = \frac{2}{5}k$$

$$2k + 3y - 4\left(\frac{2k}{5}\right) = 0$$

$$3y = \frac{8k}{5} - 2k$$

$$3y = -\frac{2k}{5}$$

$$y = -\frac{2k}{15}$$

$$x : y : z$$

$$k : -\frac{2}{15}k : \frac{2}{5}k$$

22. The system of equations

$$\alpha x + y + z = \alpha - 1$$

$$x + \alpha y + z = \alpha - 1$$

$$x + y + \alpha z = \alpha - 1 \quad \text{has no solution. Find } \alpha.$$

यदि समीकरणों के निकाय

$$\alpha x + y + z = \alpha - 1$$

$$x + \alpha y + z = \alpha - 1$$

$$x + y + \alpha z = \alpha - 1 \quad \text{का कोई हल नहीं हो, तो } \alpha \text{ का मान ज्ञात कीजिये।}$$

Ans. -2

Sol. $D = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$

$$\alpha(\alpha^2 - 1) - 1(\alpha - 1) + 1(1 - \alpha) = 0$$

$$\alpha^3 - \alpha - \alpha + 1 + 1 - \alpha = 0$$

$$\alpha^3 - 3\alpha + 2 = 0$$

$$\alpha^2(\alpha - 1) + \alpha(\alpha - 1) - 2(\alpha - 1) = 0$$

$$(\alpha - 1)(\alpha^2 + \alpha - 2)$$

$$(\alpha - 1)(\alpha^2 + 2\alpha - \alpha - 2)$$

$$(\alpha - 1)(\alpha(\alpha + 2) - 1(\alpha + 2))$$

$$\alpha = 1 \quad \alpha = -2$$

23. Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$ where x, y, z are not all zero, prove that $a^2 + b^2 + c^2 + 2abc = 1$.

दिया गया है $x = cy + bz$; $y = az + cx$; $z = bx + ay$ जहाँ x, y, z सभी शून्य नहीं हैं, सिद्ध कीजिये कि $a^2 + b^2 + c^2 + 2abc = 1$.

Ans.

Sol. $x - cy - bz = 0$

$$cx - y + az = 0 \quad x \neq 0; y \neq 0, z \neq 0$$

$$bx + ay - z = 0$$

these system is homogeneous

$$\text{so } \Delta_x = \Delta_y = \Delta_z = 0$$

and at $\Delta = 0 \rightarrow$ system have non zero solution.

$$\Delta = \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

$$1 - a^2 + c(-c - ab) - b(ac + b) = 0$$

$$1 - a^2 - b^2 - c^2 - abc - abc = 0$$

$$a^2 + b^2 + c^2 + 2abc = 1$$

24. For what values of p , the equations : $x + y + z = 1$; $x + 2y + 4z = p$ and $x + 4y + 10z = p^2$ have a solution? Solve them completely in each case.

च के किन मानों के लिये समीकरणों $x + y + z = 1$; $x + 2y + 4z = p$ तथा $x + 4y + 10z = p^2$ का हल होगा ? प्रत्येक मान के लिये इन्हें हल भी कीजिये।

Ans. $x = 1 + 2K, y = -3K, z = K$, when $p = 1$; $x = 2K, y = 1 - 3K, z = K$ when $p = 2$; where $K \in \mathbb{R}$

Sol.

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix}$$

$$D = 1(20 - 16) - 1(10 - 4) + 1(4 - 2)$$

$$D = 4 - 6 + 2 = 0$$

$$D_1 = D_2 = D_3 = 0$$

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ p & 2 & 4 \\ p^2 & 4 & 10 \end{vmatrix}$$

$$D_1 = 1(20 - 16) - 1(10p - 4p^2) + 1(4p - 2p^2) = 0$$

$$D_1 = 4 - 10p + 4p^2 + 4p - 2p^2 = 0$$

$$= 2p^2 - 6p + 4 = 0$$

$$= p^2 - 3p + 2 = 0$$

$$p^2 - 2p - p + 2 = 0$$

$$p(p - 2) - 1(p - 2) = 0$$

$$(p - 1)(p - 2) = 0$$

$$p = 1, 2$$

$$\text{for } p = 1 \quad x + y + z = 1$$

$$x + 2y + 4z = 1$$

$$x + 4y + 10z = 1$$

$$\text{for } p = 2 \quad x + y + z = 1 \quad z = k$$

$$x + 2y + 4z = 2 \quad y = 1 - 3k$$

$$x + 4y + 10z = 4 \quad x = 2k$$

$$\text{let } z = k$$

$$x + y = 1 - k$$

$$x + 2y = 1 - 4k$$

$$- \quad - \quad - \quad +$$

$$-y = 3k$$

$$y = -3k$$

$$x - 3k + k = 0$$

$$x - 2k = 1$$

$$x = 2k + 1$$

25. If the following system of equations $(a-t)x + by + cz = 0$, $bx + (c-t)y + az = 0$ and $cx + ay + (b-t)z = 0$ has non-trivial solutions for different values of t , then show that we can express product of these values of t in the form of determinant.

यदि समीकरणों के निकाय $(a-t)x + by + cz = 0$, $bx + (c-t)y + az = 0$ तथा $cx + ay + (b-t)z = 0$ के t के विभिन्न मानों के लिये अनिर्णयक हल है, तो दशाईये की t के सभी मानों के गुणन को एक सारणिक के रूप में प्रदर्शित किया जा सकता है।

Ans.
$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Sol. $(A-t)x + by + cz = 0$

$$bx + (c-t)y + az = 0$$

$$cx + ay + (b-t)z = 0$$

$$D = \begin{vmatrix} a-t & b & c \\ b & c-t & a \\ c & a & b-t \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{vmatrix} a+b+c-t & a+b+c-t & a+b+c-t \\ b & c-t & a \\ c & a & b-t \end{vmatrix} = 0$$

$$(a+b+c-t) \begin{vmatrix} 1 & 1 & 1 \\ b & c-t & a \\ c & a & b-t \end{vmatrix} = 0$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 - C_1$$

$$(a+b+c-t) \begin{vmatrix} 1 & 0 & 0 \\ b & c-t-b & a-b \\ c & a-c & b-t-c \end{vmatrix} = 0$$

$$(a + b + c - t) ((c - t - b)(b - t - c) - (a - c)(a - b)) = 0$$

$$(a + b + c - t)(bc - ct - c^2 - tb + t^2 + tc - b^2 + bt + bc - a^2 + ab + ac - bc) = 0$$

$$(a + b + c - t)(t^2 - a^2 - b^2 - c^2 + ab + bc + ca) = 0$$

$$t = a + b + c$$

$$t^2 = a^2 + b^2 + c^2 - ab - bc - ca$$

$$t^3 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 26 to 28

Consider the determinant $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$

M_{ij} = Minor of the element of i^{th} row and j^{th} column

C_{ij} = Cofactor of the element of i^{th} row and j^{th} column

माना सारणिक $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$

M_{ij} = i^{th} पक्ति तथा j^{th} स्तम्भ के अवयव का उपसारणिक

C_{ij} = i^{th} पक्ति तथा j^{th} स्तम्भ के अवयव का सहखण्ड

26. Value of $b_1 \cdot C_{31} + b_2 \cdot C_{32} + b_3 \cdot C_{33}$ is :

$b_1 \cdot C_{31} + b_2 \cdot C_{32} + b_3 \cdot C_{33}$ का मान होगा :

- (A) 0 (B) Δ (C) 2Δ (D) Δ^2

Ans. A

Sol. On multiplying elements of any row with co-factor of corresponding elements of other row then added then the resultant sum will be zero.

27. If all the elements of the determinant are multiplied by 2, then the value of new determinant is :

यदि सारणिक के सभी अवयवों को 2 से गुणा किया जाये तो नये सारणिक का मान होगा :

- (A) 0 (B) 8Δ (C) 2Δ (D) $2^9\Delta$

Ans. B

Sol. If all the elements of a det. are multiplied by k then value of that det. gets multiplied by k^n where n is the order of det.

28. $a_3 M_{13} - b_3 M_{23} + d_3 M_{33}$ is equal to :

$a_3 M_{13} - b_3 M_{23} + d_3 M_{33}$ का मान होगा :

- (A) 0 (B) 4Δ (C) 2Δ (D) Δ

Ans. D

Sol. By multiplying elements of a row with corresponding cofactors and then added, resultant sum will be equal to the value of det.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

29. The set of equations $x - y + 3z = 2$, $2x - y + z = 4$, $x - 2y + \alpha z = 3$ has :

- (A) unique solution only for $\alpha = 0$ (B) unique solution for $\alpha \neq 8$
(C) infinite number of solutions for $\alpha = 8$ (D) no solution for $\alpha = 8$

समीकरणों का समुच्चय $x - y + 3z = 2$, $2x - y + z = 4$ तथा $x - 2y + \alpha z = 3$ के पास :

- (A) अद्वितीय हल केवल $\alpha = 0$ के लिये होगा (B) अद्वितीयक हल केवल $\alpha \neq 8$ के लिये होगा
(C) $\alpha = 8$ के लिये अनन्त हल होगा (D) $\alpha = 8$ के लिये कोई हल नहीं है

Ans. BD

Sol. $D = \alpha - 8 \Rightarrow$ (B) ; If $\alpha = 8$, $D_x = 2\alpha - 14 \Rightarrow$ no solution

30. If A_{11}, A_{12}, A_{13} are the cofactors of the elements of the first row of the determinant $A = (a_{ij})$ and Δ is the value of the determinant then :

यदि A_{11}, A_{12}, A_{13} सारणिक $A = (a_{ij})$ के प्रथम पंक्ति के अवयवों के सहखण्ड है, एवं Δ सारणिक का मान हैं, तो :

- (A) $a_{21} A_{11} + a_{22} A_{12} + a_{23} A_{13} = 0$ (B) $\alpha_{11} A_{11} + \alpha_{12} A_{12} + \alpha_{13} A_{13} = \Delta$
(C) $\alpha_{11} A_{11} - \alpha_{12} A_{12} + \alpha_{13} A_{13} = \Delta$ (D) $a_{31} A_{11} + a_{32} A_{12} + a_{33} A_{13} = 0$

Ans. ABD

Sol. By property

31. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ if :

(A) a, b, c are in A.P.

(B) a, b, c are in G.P.

(C) α is a roots of $ax^2 + bx + c = 0$

(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$

सारणिक $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ है, यदि :

(A) एइएब स.श्रे. में है

(B) एइएब गु.श्रे. में है

(C) α समीकरण $ax^2 + bx + c = 0$ का मूल है

(D) $x^2 + 2bx + c = 0$ का गुणनखण्ड $(x - \alpha)$ है

Ans. BD

Sol. $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$

$$R_3 \rightarrow R_3 - (\alpha R_1 + R_2)$$

$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ 0 & 0 & -a\alpha^2 - 2b\alpha - c \end{vmatrix} = 0$$

$$(-a\alpha^2 - 2b\alpha - c)(ac - b^2)$$

$$(a\alpha^2 + 2b\alpha + c)(ac - b^2) = 0$$

$$a\alpha^2 + 2b\alpha + c = 0$$

α root of equation.

$$ac = b^2 \text{ G.P.}$$

32. The parameter, on which the value of $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ depend upon, is :

वह प्राचाल जिस पर $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ का मान निर्भर करता है :

(A) a

(B) p

(C) d

(D) x

Ans. ACD

Sol. $\sin((p+d)x - px) - a(\sin((p+d)x - (p-d)x) + a^2(\sin(px - (p-d)x))$
 $\sin dx - a \sin(2dx) + a^2 \sin dx$

MATCH THE COLUMN TYPE QUESTION(S)

33. Match the following Columns :

Column-I

Column-II

- (A) Let $|A| = |a_{ij}|_{3 \times 3} \neq 0$. Each element a_{ij} is multiplied by k^{i-j} . Let $|B|$ be the resulting Determinant, where $k_1|A| + k_2|B| = 0$. Then $k_1 + k_2 =$
- (B) The maximum value of a third order determinant each of its entries are ± 1 equals

(P) 0

(Q) 4

(C) If $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$ then

(R) 1

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma =$$

(D) $\begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = Ax + B$ where A and B are

(S) 2

determinants of order 3. Then $A + 2B =$

निम्न Columns को सुमेलित कीजिए :

Column-I

Column-II

- (A) माना $|A| = |a_{ij}|_{3 \times 3} \neq 0$ प्रत्येक अवयवों a_{ij} , k^{i-j} से गुणा किया जाये तो यदि $|B|$ परिणामी सारणिक है, तो यदि $k_1|A| + k_2|B| = 0$ तो $k_1 + k_2 =$
- (B) एक क्रम तीन के सारणिक का प्रत्येक अवयव ± 1 हो सकता है, तो सारणिक का अधिकतम मान होगा

(P) 0

(Q) 4

(C) यदि $\begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$ तो

(R) 1

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma =$$

(D) $\begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = Ax + B$ जहाँ A तथा B क्रम 3 के

(S) 2

सारणिक है, तो $A + 2B =$

Ans. (A) P (B) Q (C) R (D) P

Sol. (i) $B = \begin{vmatrix} 1 - k^{-1} - \frac{1}{k^2} & k - 1 - \frac{1}{k} & k^2 - k - 1 \\ k^2 - k - 1 & k - 1 - \frac{1}{k} & k^2 - k - 1 \\ k^2 - k - 1 & k - 1 - \frac{1}{k} & k^2 - k - 1 \end{vmatrix}$

$$B = \frac{1}{k^2 \cdot k} \begin{vmatrix} k^2 - k - 1 & k - 1 - \frac{1}{k} & k^2 - k - 1 \\ k^2 - k - 1 & k - 1 - \frac{1}{k} & k^2 - k - 1 \\ k^2 - k - 1 & k - 1 - \frac{1}{k} & k^2 - k - 1 \end{vmatrix}$$

$$B = \begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix}$$

$$|B| = |A|$$

$$|A| - |B| = 0$$

$$k_1 = 1 \quad k_2 = -1$$

$$k_1 + k_2 = 0$$

$$(iii) \begin{vmatrix} 1 & \cos \alpha & \cos \beta \\ \cos \alpha & 1 & \cos \gamma \\ \cos \beta & \cos \gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos \alpha & \cos \beta \\ \cos \alpha & 0 & \cos \gamma \\ \cos \beta & \cos \gamma & 0 \end{vmatrix}$$

$$(1 - \cos^2 \gamma) - \cos \alpha (\cos \alpha - \cos \beta \cos \gamma) + \cos \beta (\cos \alpha \cos \gamma - \cos \beta)$$

$$= 0 - \cos \alpha (0 - \cos \beta \cos \gamma) + \cos \beta (\cos \alpha \cos \gamma)$$

$$1 - \cos^2 \gamma - \cos^2 \alpha + \cos \alpha \cos \beta \cos \gamma + \cos \alpha \cos \beta \cos \gamma - \cos^2 \beta = 2 \cos \alpha \cos \beta \cos \gamma$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$(d) \begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x-3 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$$

$$R_2 \rightarrow R_2 - (R_1 + R_3)$$

$$\begin{vmatrix} x^2 + x & x+1 & x-2 \\ -4 & 0 & 0 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$$

$$4((x+1)(2x-1) - (2x-1)(x-2)) = Ax + B$$

$$4(2x^2 - x + 2x - 1 - 2x^2 + 4x + x - 2) = Ax + B$$

$$4(6x - 3) = Ax + B$$

$$24x - 12 = Ax + B$$

$$A = 24, B = -12$$

$$A + 2B$$

$$24 - 2 \times 12 \Rightarrow 0$$